Technical Note — Unified Overlay Runner (Indigenous + MPO)

1) Background & Purpose

The Unified Overlay Runner applies counterfactual policy overlays (Indigenous and MPO) to MPI projects. It does not re-estimate Bayes or Cox models. Instead, it simulates probability shifts and expected value uplifts.

2) Inputs

- Required: blended_prob, project_cost, t0_orig
- Optional: EV_disc, reporting_years, Indigenous flag column
- Recommended: province, sector, group, FOAK, cleantech, cost_band for grouping

3) Method

Indigenous overlay:

Applies Δ years (default –1) and hazard multiplier to flagged projects.

MPO overlay:

Caps approvals duration at 2 years; applies odds scaling.

Combined overlay:

Sequential application (default: MPO → Indigenous).

Expected Value (EV):

EV = probability × project_cost (millions CAD). If EV_disc present, uses discounted project cost.

4) Outputs

- {scenario}_overlay_results.csv — per-project outputs
- {scenario}_summary_{group}.csv — summaries by province, sector, group, FOAK, cleantech, cost_band
- {scenario}_report.md — concise Markdown roll-up

5) Replication Procedure

- Prepare input CSV with required fields
- Run Colab-ready script unified_overlay_runner_colab_revised_v2.py
- Check per-project and grouped outputs
- Archive input/output and report files

6) QA / Audit Checks

- Confirm Δ probabilities within [0,1]
- Check crossers (blended <0.5 to ≥0.5) and cross-outs
- Validate years_in_pipeline = 2024 - t0_orig + 1
- Confirm EV_disc used when available

7) Limitations

Counterfactual only; no re-estimation; stylized parameters; outputs show distributional shifts, not forecasts.